Math 199, Spring 2026
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**Week 11 - Equivalence relations**

**Definition:** Let $R \subset A \times A$ be a relation on the set $A$.

1. The relation $R$ is **symmetric** if for all $(a, b) \in R$, we also have $(b, a) \in R$.
2. The relation $R$ is **transitive** if for all $a, b, c \in A$, if $(a, b) \in R$ and $(b, c) \in R$, then we also have $(a, c) \in R$.
3. The relation $R$ is **reflexive** if for all $a \in A$, we have $(a, a) \in R$.
4. The relation $R$ is an **equivalence relation** if $R$ is symmetric, transitive, and reflexive.
5. If $R$ is an equivalence relation on $A$, then for each $a \in A$, we define its **equivalence class** as the set $[a]_R = \{b \in A : (a, b) \in R\}$.
6. If $R$ is an equivalence relation on $A$, then we let $A/R$ denote the set of all equivalence classes of elements of $A$, i.e.
$$A/R = \{[a] \subset A : a \in A\}.$$

1) Let $R_n \subset \mathbb{Z} \times \mathbb{Z}$ be the relation defined by

$$R_n = \{(x, y) \in \mathbb{Z}^2 : x \equiv y \mod n\}.$$

(a) Prove that $R_n$ is an equivalence relation.

(b) Prove that $(\mathbb{Z}/R_n, +_n, [0]_{R_n})$ is an abelian group, where $[x]_{R_n} +_n [y]_{R_n} = [x+y]_{R_n}$. (Note: you also have to justify that $+_n$ is well-defined.)

(c) Last week, we encountered a group $\mathbb{Z}/n = \{0, ..., n-1\}$, under addition mod $n$. Show that there is a group isomorphism, $\mathbb{Z}/R_n \to \mathbb{Z}/n$. (i.e. define a function, show it is well-defined, show it is a group homomorphism, and show it is an isomorphism).

2) Let $Q \subset \mathbb{R} \times \mathbb{R}$ be the relation defined by

$$Q = \left\{ (a,b) \in \mathbb{R}^2 : \frac{a}{b} \in \mathbb{Q} \right\}.$$

(a) Prove that $Q$ is transitive.

(b) Is $Q$ symmetric and/or reflexive? Prove why or why not.

(c) Let $Q_0 \subset Q \subset \mathbb{R}^2$ be the set of all $(a, b) \in Q$ such that $a \neq 0$. Is $Q_0$ an equivalence relation? Prove your answer.

3) Let $G = (G, \cdot, e)$ be a group and define a relation $C$ on $G$ by

$$C = \{(a, b) \in G^2 : \exists g \in G \text{ with } a = gbg^{-1}\}.$$

(a) Prove that $C$ is an equivalence relation.

(b) Prove that a group $G$ is abelian if and only if the function $f : G \to G/C$ defined by $f(a) = [a]_C$ is a bijection.

(c) Does the the group operation on $G$ always define a well-defined group operation on the set $G/C$? Prove why or why not.

4) Recall that functions are special types of relations. Remind yourself of the definition of when a relation is a function. In this problem we explore some examples of how the propetry of being a function relates with the other properties above.

(a) Show that there exists a relation on $\mathbb{R}$ that is reflexive, symmetric, and a function.

(b) Show that there exists a relation on $\mathbb{R}$ that is symmetric, is a function, and is not reflexive.

(c) Prove that there exists a relation on $\mathbb{R}$ that is both transitive and a function that is neither the identity function, nor a constant function.

(d) Prove that there exists exactly one equivalence relation on $\mathbb{R}$ that is also a function.